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Started on	2024-04-18 00:30
State	Finished
Completed on	2024-04-18 03:57
Time taken	3 hours 27 mins
Grade	10.00 out of 10.00 (100%)

Question 1

Complete
Mark 2.00 out of 2.00

Using the principle of mathematical induction, we are going to show that $3^n - 1$ is a multiple of 2.

Please write the predicate $P(n)$ and the basis step of the proof.

Attach your answer below.

IMG_20240418_035442.jpg

Comment:

Question 2

Complete
Mark 3.00 out of 3.00

Using the principle of mathematical induction, we are going to show that $3^n - 1$ is a multiple of 2.

Please write the predicate $P(n)$ and the inductive step of the proof.

Attach your proof below.

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Comment:

Question 3

Complete
Mark 2.00 out of 2.00

Explain how the recursive bubble sort works to sort 4, 3, 2, 5 into increasing order.

Attach your answer below.

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Comment:

Question 4

Complete

Mark 2.00 out of 2.00

Use a merge sort construct a tree to sort $4, 3, 2, 5, 1, 8, 7, 6$ into increasing order.

Attach your answer below.

 IMG_20240418_035453.jpg

Comment:

Question 5

Correct

Mark 1.00 out of 1.00

Find big-O estimate for the following recurrence relation using the Master theorem $T(n) = 8T(n/4) + 5n^2$.

Select one:

☒ $O(n^2)$ ✓

☐ $O(n^2 \log n)$

☐ $O(n^{3/2} \log n)$

☐ $O(n^{3/2})$

Your answer is correct.

The correct answer is: $O(n^2)$